

ZHAOLI CAO

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EDUCATION

The University of Hong Kong

Master of Finance in Financial Technology | GPA: 4.0/4.0

Hong Kong, China

2025.09 – 2026.11

- **Honors:** Vice President of the HKU Elite Club (Fintech)

Fudan University

Bachelor of Science in Mathematics (Minor: Economics & Finance) | GPA: 3.76/4.0

Shanghai, China

2020.09 – 2025.06

- **Honors:** WorldQuant Challenge Gold Medal (2025), First-Class Scholarship of Fudan University(2024), IMC Prosperity3 Global Top 2% (2025), Kaggle DRW Crypto Market Prediction Top 10% (2025), MCM Meritorious Winner (2023)

PROFESSIONAL EXPERIENCE

NonConvex Technology, Quantitative Researcher(HFT: 35B average daily volume) Shanghai, Mar 2026 – Present

Conducted proprietary HFT research on equity index futures, focusing on microstructure alpha and execution optimization

- **Low-Latency Execution:** Built Rust-based execution modules with queue-reactive order slicing and impact-aware execution

Matrixport, Quantitative Trader (AUM: \$10B+ in digital assets)

Hong Kong, Sep 2025 – Dec 2025

Conducted crypto microstructure research, focusing on L2 signal extraction and multi-frequency factor modeling.

- **Market Making Strategy Research (Crypto):** Studied inventory-aware optimal quoting in a limit order book, adjusting bid-ask spreads and order skew using L2 order book dynamics and funding-rate signals to improve execution quality in backtests.
- **Microstructure & Regime Detection:** Based on L2 order book and funding rate data, extracted market microstructure features and employed a Hidden Markov Model to identify latent market states, enabling dynamic signal adaptation.
- **Multimodal Temporal Modeling:** Optimized an end-to-end time-series modeling framework that integrates L2 order book data with NLP-based sentiment features, utilizing LSTM/Transformer to capture market dynamics.

Geek Union Fund, Quantitative Research

Shanghai, Jun 2025 – Aug 2025

Responsible for factor mining on daily and hourly CTA data, as well as multi-window factor training and portfolio construction.

- **Factor Mining:** Conducted daily/hourly CTA factor research via multi-window training, dynamic smoothing, and IC–return filtering to build a library of 80+ signals. Applied regime-sensitive sampling to improve robustness across market states
- **Operator Development:** Refactored core signal modules, achieving ~77% runtime improvement via Numba JIT and pybind11-based C++ extensions with SIMD (AVX2). Integrated multi-horizon operators, cross-sectional allocation, and risk-budget weighting to speed up tuning and portfolio construction
- **Loss & Statistical Regularization:** Designed a robust loss (winsorization, Huber, directional penalties) and applied Newey–West adjustments to address heteroskedasticity and autocorrelation in factor evaluation
- **Rule-Based Features & Portfolio:** Extracted rule-based K-line features (IC up to 0.12) and built multi-factor portfolios using SGD-based allocation + KMeans, achieving Sharpe 2.64 / Calmar 4.15 out-of-sample

KR Capital, Quantitative Research

Shanghai, Jan 2025 – May 2025

Responsible for CTA strategy and stock strategy research, backtesting, and deployment in simulation and live trading

- **Cross-Sectional Futures Alpha:** Built minute-level multi-factor long–short strategies using roll yield, basis momentum, and other high-frequency features. Applied XGBoost and Newey–West residual adjustments, achieving Sharpe 1.78, mdd 9.6%
- **High-Frequency Statistical Arbitrage:** Built a latency-aware stat arb strategy with dynamic hedge ratios (rolling GLS), accounting for execution frictions and validated out-of-sample, achieving Sharpe 3.45 after accounting for transaction costs
- **Global Macro Rotation:** Built a regime-based ETF allocation strategy using GMM (EM), achieving live Sharpe 2.35

PROJECT EXPERIENCE

Self-Built End-to-End Quant Framework (A-share / US / Crypto, 50K LOC)

Mar 2026

Built prod-grade system (Rust FE + Polars + Numba JIT), signals cross-validated vs. industry benchmarks (QuantConnect)

- PIT 3-layer pipeline (announce_date constraint) + BacktestClock physical truncation, eliminating look-ahead bias
- Unified Delta2Planner(e exec-decoupled via protocol)+stateful matcher modeling fills, impact & capacity constraints
- Barra CNE6 (mcap-weighted WLS+Newey–West+Ledoit–Wolf shrinkage) + 2-layer risk (port opt + per-order RiskGuard)

IMC Trading Prosperity3 High-Frequency Trading Competition (Global Top 2%)

Apr 2025

Served as team leader, overseeing case studies and strategy development; led quantitative strategy design and manual trading

- Strategies included: order book market making (spread & inventory control), ETF premium arbitrage (intra-day dislocations), options mispricing via Black–Scholes/IV bias, cross-asset relative value, and signal-based momentum following.

Bitcoin Price Prediction and Quantitative Trading Strategy Based on Machine Learning

Feb 2025 - May 2025

Constructed bitcoin price prediction model and designed optimized quantitative trading strategy

- Built BTC prediction and trading framework(feature engineer+Bayesian optimization+walk-forward backtest), Sharpe 2.70

SKILLS

Programming: Rust(proficient), C++ (advanced: STL, RAI), Python (advanced: NumPy, Pandas, Pytorch), Linux, R, Matlab

Language: Chinese (native), English (fluent, TOEFL: 109 (Reading: 30; Listening: 29; Speaking: 25; Writing: 25))

曹肇立

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教育背景

- 香港大学，金融科技 | GPA(4.0/4.0)** 香港, 2025.09 - 2026.11
- 荣誉职位: 香港大学精英协会 金融科技协会副会长
- 复旦大学，数学与应用数学（辅修专业：经济学&金融学） | GPA(3.76/4.0)** 上海, 2020.09 - 2025.06
- 荣誉奖项: WorldQuant金牌(2025)、Kaggle DRW Crypto Market Prediction 全球前10%(2025)、高频交易比赛IMC Prosperity全球前2%(2025)、美赛M奖(2023)、复旦大学专业奖学金

实习经历

- 非凸科技，高频量化研究（自营交易，日均成交约350亿元）** 上海, 2026.03 – 至今
- 负责股指期货高频策略研究，涵盖微观结构信号挖掘、执行算法优化及低延迟交易系统开发
- 执行优化: Rust重构低延迟执行模块, 实现queue-reactive拆单与impact-aware execution, 优化成交概率/冲击
- Matrixport，量化交易（数字资产管理规模: 100亿美元以上）** 香港, 2025.09 – 2025.12
- 负责Crypto市场的量化研究与交易，高频因子挖掘、验证与优化
- 做市策略建模: 围绕限价订单簿下的库存约束做市问题，研究inventory-aware最优报价机制，引入成交强度与库存惩罚项建模，结合 L2 订单簿动态与资金费率信号，对bid - ask spread与挂单倾斜进行自适应调整
 - 微观结构与状态识别: 基于 L2 订单簿与资金费率数据提取市场微观结构特征，采用隐马尔可夫模型HMM识别潜在市场状态，按状态后验概率对进出场阈值与因子暴露进行加权调整，策略 Sharpe 提升约 23%
 - 多模态时序建模: 构建端到端策略，采用双塔结构与门控融合机制，结合状态条件化与表示对齐优化，融合 L2 订单簿序列与 NLP 情绪特征，通过 LSTM / Transformer 学习状态条件化信号，测试集Sharpe2.77
- 积优基金，量化研究** 上海, 2025.06 – 2025.08

- CTA日频及小时频因子挖掘、因子多窗口期训练及组合**
- 单因子训练: 多窗口训练，结合IC-收益联合筛选80+因子，并以状态分层建模检验异质市场下的稳健性
 - 损失函数及统计修正: 设计自定义损失函数，结合95分位数winsorize、Huber Loss与方向一致性惩罚，增强异常值鲁棒性与信号稳健性；在因子评估中引入Newey - West回归修正，应对异方差与自相关问题
 - 规则类与多因子组合: 基于K线模式提取规则类特征，单因子IC值0.12；采用基于SGD的Base分钱方法与Kmeans聚类构建多因子组合，五因子组合在超样本中实现Sharpe 2.64、Calmar 4.15
 - 框架优化: 重构模块, 基于Numba, PyTorch与SIMD(AVX2)实现77%性能提升并引入多周期预测&风险预算

- KR Capital，量化研究** 上海, 2025.01 – 2025.05
- 负责期货截面策略和股票量化策略的研究回测及模拟盘实盘部署
- 期货截面Alpha: 分钟级多空组合(期限结构/基差), XGBoost+滚动筛选(NW修正), Sharpe 1.78 / MDD 9.6%
 - 全球宏观轮动策略: 利用GMM+EM聚类划分市场状态并动态调整ETF配置，实盘部署至今Sharpe 2.35
 - 配对交易(TSLA-NVDA): 构建考虑延迟的统计套利策略(滚动GLS)，纳入执行摩擦,样本外验证Sharpe 3.45

项目经历

- 自建全流程量化交易框架（A股/美股/加密货币，5万行个人项目）** 上海, 2026.03
- 基于Rust内核因子引擎+Polars+Numba JIT构建，全链路信号与聚宽/RQAlpha交叉验证对齐
- 数据与防泄露: PIT三层数据管道(announce_date约束)+ BacktestClock物理截断未来信息, 杜绝look-ahead bias
 - 回测/实盘一致: 统一Delta2Planner调仓核心(仅替换执行层), 状态撮合引擎建模累计成交、冲击成本与资金
 - 风控: Barra CNE6(市值加权WLS+Newey-West+Ledoit-Wolf收缩)，双层风控分离(组合级优化 + 逐单级RiskG)
- IMC Prosperity3高频交易竞赛（全球前2%）** 上海, 2025.04
- 担任队长，带领团队学习往年案例，分析数据，构建并优化策略，并负责开发量化策略及所有手动交易
- 构建基于订单簿深度与价格波动的做市报价模型、利用ETF市场价与组件净值间价差的跨品种套利机制、结合Black-Scholes期权定价与隐含波动率偏差的定价套利模型、基于协整与价差Z-score构造的跨资产价差回归策略
- 基于机器学习的比特币价格预测与量化交易策略研究（优秀毕业论文）** 上海, 2025.02 – 2025.05
- 构建比特币价格预测模型，并设计优化的量化交易策略
- 构建多模型比特币预测与量化策略，引入特征工程与贝叶斯优化，结合滚动回测与成本建模，Sharpe2.70

个人技能

编程语言: Rust (熟练), C++ (精通: STL, RAI), Python (精通: NumPy, Pandas, SciPy, Pytorch), Linux, R, Matlab

语言能力: 中文(母语)、英文(流利, TOEFL: 109; 阅读: 30, 听力: 29, 口语: 25, 写作: 25)